

exp3 — outcome by setup and heterogeneity condition (K = 10 unless noted)
cell state = majority of runs · ratio = Kaplan-Meier median first crossing / matched IID

	A: quad-MLP, mod 97, GD	B: transformer, mod 113	C: transformer, S ₅ (withheld)	D: quad-MLP, S ₅	E: MLP, MNIST
IID	✓ 5/5 crossed memorised 5/5 · peak train 100%	✓ 3/3 crossed memorised 3/3 · peak train 100%	✓ 3/3 crossed memorised 3/3 · peak train 100%	✓ 3/3 crossed memorised 3/3 · peak train 100%	✓ 3/3 crossed memorised 3/3 · peak train 100%
Dirichlet 1000	✓ 3/3 crossed 1.0× memorised 3/3 · peak train 100%	✓ 3/3 crossed 1.5× memorised 3/3 · peak train 100%	✓ 3/3 crossed 1.0× memorised 3/3 · peak train 100%	✓ 3/3 crossed 0.6× memorised 3/3 · peak train 100%	✓ 3/3 crossed 0.8× memorised 3/3 · peak train 100%
Dirichlet 10	✓ 3/3 crossed 1.0× memorised 3/3 · peak train 100%	✓ 3/3 crossed 1.3× memorised 3/3 · peak train 100%	✓ 3/3 crossed 1.4× memorised 3/3 · peak train 100%	✓ 3/3 crossed 0.8× memorised 3/3 · peak train 100%	✓ 3/3 crossed 0.9× memorised 3/3 · peak train 100%
Dirichlet 1	✓ 3/3 crossed 1.0× memorised 3/3 · peak train 100%	✓ 3/3 crossed 0.8× memorised 3/3 · peak train 100%	✓ 3/3 crossed 1.6× memorised 3/3 · peak train 100%	⦿ 0/3 crossed memorised 3/3 · peak train 100%	✓ 3/3 crossed 0.7× memorised 3/3 · peak train 100%
Dirichlet 0.5	✓ 5/5 crossed 1.0× memorised 5/5 · peak train 100%	✓ 3/3 crossed 0.9× memorised 3/3 · peak train 100%	✓ 3/3 crossed 1.6× memorised 3/3 · peak train 100%	⦿ 0/3 crossed memorised 3/3 · peak train 100%	✓ 3/3 crossed 0.8× memorised 3/3 · peak train 100%
Dirichlet 0.1	✓ 3/3 crossed 1.0× memorised 3/3 · peak train 100%	✗ 0/3 crossed memorised 0/3 · peak train 31%	✓ 2/3 crossed 3.5× memorised 0/3 · peak train 97% · mixed	✗ 0/3 crossed memorised 0/3 · peak train 60%	✓ 3/3 crossed 0.8× memorised 3/3 · peak train 100%
Dirichlet 0.01	✓ 3/3 crossed 1.1× memorised 3/3 · peak train 100%	✗ 0/3 crossed memorised 0/3 · peak train 20%	✗ 0/3 crossed memorised 0/3 · peak train 59%	✗ 0/3 crossed memorised 0/3 · peak train 5%	not run
operand	✓ 3/3 crossed 0.9× memorised 3/3 · peak train 100%	✓ 2/3 crossed 9.7× memorised 2/3 · peak train 100% · mixed	✓ 3/3 crossed 1.2× memorised 3/3 · peak train 100%	✗ 0/3 crossed memorised 0/3 · peak train 98%	not run
label	✓ 3/3 crossed 1.1× memorised 3/3 · peak train 100%	✗ 0/3 crossed memorised 0/3 · peak train 15%	✗ 0/3 crossed memorised 0/3 · peak train 33%	✗ 0/3 crossed memorised 0/3 · peak train 3%	✗ 0/3 crossed memorised 0/3 · peak train 97%
✓ grokked ⦿ memorised only ✗ not memorised (train < 99%)					